

CHAPTER 8: SELECTION BIAS

Foundations, Structural DAGs & Adjustment Methods

Applied Use Case: "Did the Retention Offer Actually Reduce Churn?"

Executive Overview: Chapter 8 At A Glance

Chapter 8 of *Causal Inference: What If* introduces **Selection Bias**—the second major structural barrier to causal identification alongside confounding. Unlike confounding (which arises from common causes of treatment and outcome), selection bias is caused by **conditioning on a common effect (a collider)** or its descendant.

Core Key Takeaways:

- **Structural Mechanism:** Selection bias occurs when analysis is restricted to a subset of individuals defined by a collider C or S ($A \rightarrow [C] \leftarrow Y$ or $A \rightarrow [C] \leftarrow U \rightarrow Y$).
- **Ubiquity in Research:** Affects both observational studies and randomized trials whenever post-randomization loss to follow-up, non-response, or missing data occurs.
- **Hazard Ratio Built-in Bias:** Conditioned survival past time t induces selection bias due to depletion of susceptible individuals (frailty).
- **Correction via IP Weighting:** Adjusted using Inverse Probability of Selection Weights ($W^C = 1/\Pr[C = 0 \mid A, L]$) to construct an uncensored pseudo-population.

1. The Structure of Selection Bias

Ref: Section 8.1, pp. 103–105

Selection bias is an association between treatment A and outcome Y induced by the process by which individuals are selected into the analysis dataset. Structurally, it stems from **conditioning on a common effect (a collider)** of two variables, one of which is or is caused by the treatment, and the other is or is caused by the outcome (or an unmeasured cause U of the outcome).

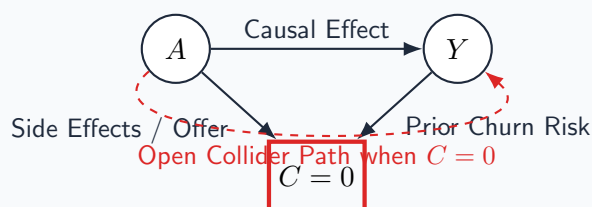


Figure 8.1: Conditioning on Collider $C = 0$ opens the non-causal path $A \rightarrow [C] \leftarrow Y$, inducing selection bias under the null.

- **CRM Churn Use Case:** Suppose a telecom company offers a retention discount (A). Receiving an offer may cause customer annoyance or contact fatigue, leading to opt-out/censoring (C). Concurrently, customers with high baseline churn risk (Y) are also more likely to opt out or drop out (C). Restricting the analysis to active, non-opt-out customers ($C = 0$) opens the collider path $A \rightarrow [C] \leftarrow Y$, creating a spurious association between the offer and churn.

- **Bias Under the Null:** Selection bias creates an association between A and Y even when the true causal effect of A on Y is strictly zero ($E[Y^{a=1}] - E[Y^{a=0}] = 0$).

2. Real-World Examples of Selection Bias

Ref: Section 8.2, pp. 105–107

Selection bias manifests across numerous business and medical scenarios depending on how data is filtered:

1. **Differential Loss to Follow-up / Informative Censoring:** Patients or customers drop out due to treatment side effects ($A \rightarrow C$) or disease severity/dissatisfaction ($U \rightarrow C, U \rightarrow Y$).
2. **Missing Data / Non-response Bias:** Restricting analysis to customers who complete voluntary feedback surveys ($S = 0$).
3. **Healthy Worker / Active Account Bias:** Evaluating workplace chemical exposures only among active workers, ignoring those who left the job due to exposure-related illness ($A \rightarrow C$).
4. **Self-Selection / Volunteer Bias:** Opt-in retention or loyalty programs where health-conscious or engaged users self-select into participation (C).
5. **Pre-Study Treatment Selection:** Assessing lifetime exposure among 50-year-olds where past treatment influenced survival/qualification prior to study entry.
6. **Case-Control Selection Bias (Fine Point 8.1):** Oversampling controls based on a condition (e.g., hip fracture) influenced by post-menopausal estrogen treatment A .

3. Selection Bias vs. Confounding & Built-in Hazard Ratio Bias

Ref: Section 8.3, pp. 107–109; Technical Point 8.1, p. 104

Structural Distinction: Confounding vs. Selection Bias

- **Confounding:** Lack of exchangeability caused by **common causes** of treatment and outcome ($A \leftarrow L \rightarrow Y$). Randomization at baseline *eliminates* confounding.
- **Selection Bias:** Lack of exchangeability caused by **conditioning on common effects** ($A \rightarrow [C] \leftarrow Y$ or $A \rightarrow [C] \leftarrow U \rightarrow Y$). Randomization at baseline *does NOT protect* against selection bias if selection occurs post-randomization.

The Built-in Selection Bias of Hazard Ratios (Technical Point 8.1): In time-to-event / survival analysis, hazard ratios conditional on survival past time t ($Y_1 = 0$) inherently condition on a collider:

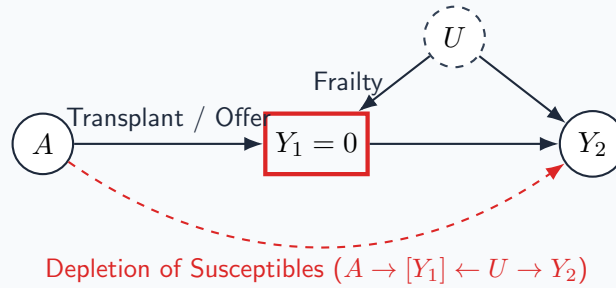


Figure 8.8: Conditioning on survival at month 1 ($Y_1 = 0$) induces frailty selection bias for mortality at month 2 (Y_2), causing hazard ratio crossing even under null direct effect.

4. Selection Bias and Censoring: Counterfactual Framework

Ref: Section 8.4, pp. 109–111

To rigorously address censoring, we frame censoring C (1 = censored, 0 = uncensored) as a **joint treatment** alongside intervention A :

- **Joint Counterfactual Outcome** $Y^{a,c=0}$: The potential outcome an individual would experience if assigned to treatment level $A = a$ and **enforced to remain uncensored** ($C = 0$).
- **Causal Contrast of Interest:**

$$\text{Causal Risk Difference} = \Pr[Y^{a=1,c=0} = 1] - \Pr[Y^{a=0,c=0} = 1]$$

- **Conceptual Insight:** Adjusting for selection bias is equivalent to adjusting for confounding of the causal effect of *censoring* C .

5. How to Adjust for Selection Bias via IP Weighting

Ref: Section 8.5, pp. 111–115

Selection bias can be corrected by calculating **Inverse Probability of Selection Weights** (W^C) for uncensored individuals ($C = 0$):

$$W^C = \frac{1}{\Pr[C = 0 \mid A, L]}$$

When both confounding ($L \rightarrow A$) and selection bias ($L \rightarrow C$) exist, the joint IP weight is the product:

$$W^{A,C} = W^A \times W^C = \frac{1}{f(A \mid L)} \times \frac{1}{\Pr[C = 0 \mid A, L]}$$

Three Identifiability Conditions for Selection Adjustment

1. **Conditional Exchangeability for Censoring:** $Y^{a,c=0} \perp\!\!\!\perp C \mid A, L$. Unmeasured factors U affecting both C and Y must be blocked by L .

2. **Positivity for Uncensoring:** $\Pr[C = 0 \mid A = a, L = l] > 0$ for all strata l where $\Pr[A = a, L = l] > 0$.

3. **Consistency:** Observed $Y = Y^{A,C=0}$ for uncensored subjects with assigned A .

Why Stratification Fails for Selection Bias

Unlike IP Weighting (which creates an uncensored pseudo-population), conventional **stratification** on L fails when L is affected by prior treatment A ($A \rightarrow L \rightarrow C \leftarrow U \rightarrow Y$). Conditioning on L opens new collider paths ($A \rightarrow [L] \leftarrow U \rightarrow Y$), creating self-inflicted selection bias.

6. Selection Without Bias (Augmented DAGs)

Ref: Section 8.6, pp. 115–118

Conditioning on a collider does **not** always cause selection bias if treatment A and secondary exposure E act through **completely independent cause-specific mortality mechanisms** (Y_A, Y_E, Y_O):

$$\Pr[Y = 0 \mid A = a, E = e] = g(e) \cdot h(a) \quad (\text{Multiplicative Survival Model})$$

When conditioning on survival ($Y = 0$), $Y_A = 0, Y_E = 0, Y_O = 0$ are simultaneously fixed, blocking the collider path between A and E .

7. Methodological Comparison across CRM & Clinical Domains

Structural Threat	Causal DAG Pattern	CRM Retention Context	Correction Method
Confounding	$A \leftarrow L \rightarrow Y$	Baseline complaint score L drives offer A & churn Y .	Standardization / W^A Weights
Selection Bias	$A \rightarrow [C] \leftarrow U \rightarrow Y$	Retention offer A causes opt-out C ; dissatisfaction U causes churn Y .	Censoring IP Weights W^C
Hazard Ratio Bias	$A \rightarrow [Y_1] \leftarrow U \rightarrow Y_2$	Depletion of high-risk customers past Month 1 alters Month 2 hazards.	Survival curves $S(t)$ / Risk Ratios
M-Bias	$U_1 \rightarrow [L] \leftarrow U_2$	Adjusting for pre-treatment collider L creates spurious path.	Exclude L from adjustment set